Machine Intelligence 1 Tutorial

10 Multi-class SVM & Support Vector Regression (SVR)

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 25/26

Multi-class SVM

- One-vs-rest (one-vs-all)
- One-vs-one (all pairs)
- Binary classification with error-correcting outputs

Support Vector Regression (SVR)

- The ε-insensitive cost function for regression
- Deriving the primal problem for ε-SVR

ν-SVR

- Motivation
- Deriving the primal problem for ν-SVR
- Deriving the Lagrangian for the primal problem of ν-SVR
  - Identifying the Lagrange multipliers
  - Re-arranging the constraints
- Optimizing ν-SVR

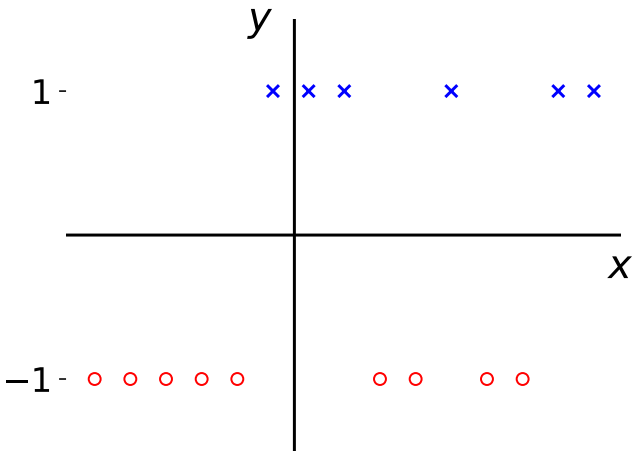
References

Multi-class SVM

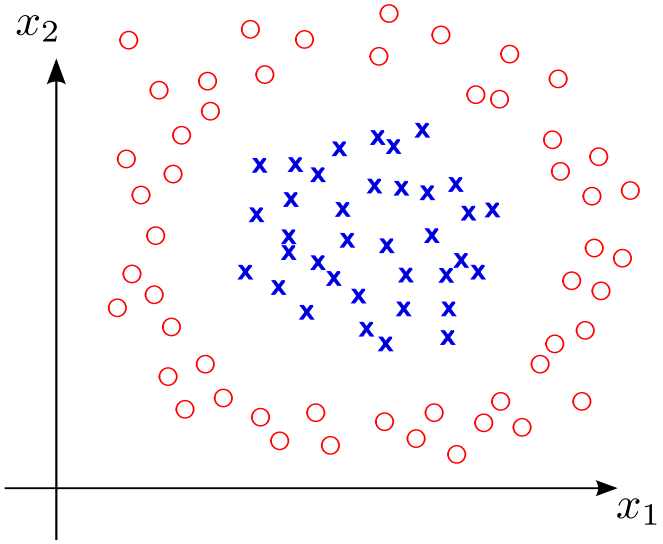
A very brief overview

The binary classification setting

Classification problems have labels with only finitely many possible values.
Example: Binary classification, $y_T \in \{-1, 1\}$:



(a) 1D input



(b) 2D input

Figure: binary classification

The multi-class classification setting

A set of $K > 2$ classes: $\{c_1, c_2, \ldots, c_K\}$

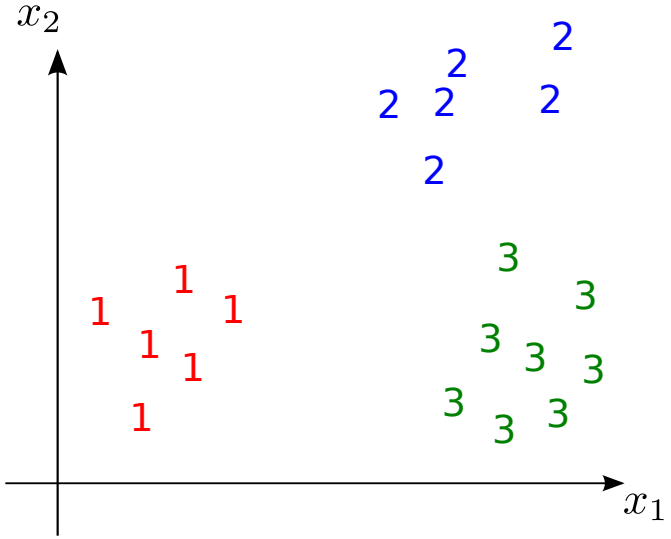


Figure: multi-class classification

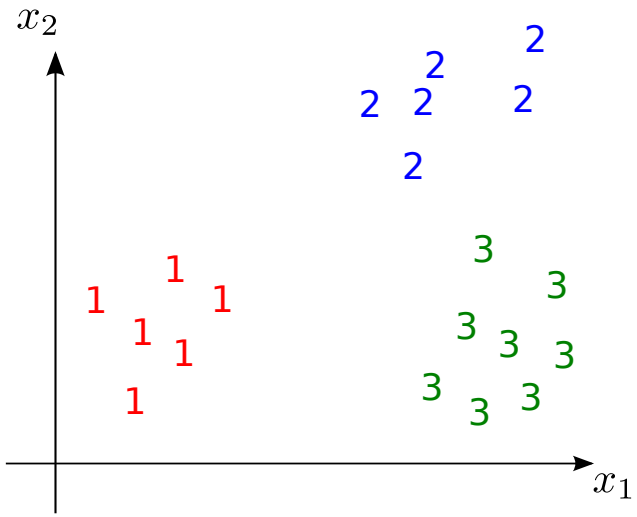
Remains an open issue for SVMs.

One-vs-rest (one-vs-all)

for each class c_k in $\{c_1, c_2, \ldots, c_K\}$
do

train a binary classifier with c_k as its **positive** class
all remaining $K-1$ classes are treated collectively as the **negative** class.

end



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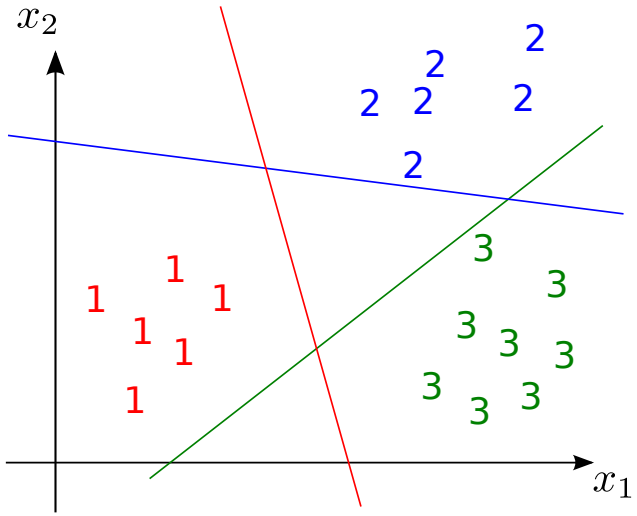

Figure: K one-vs-all classifiers

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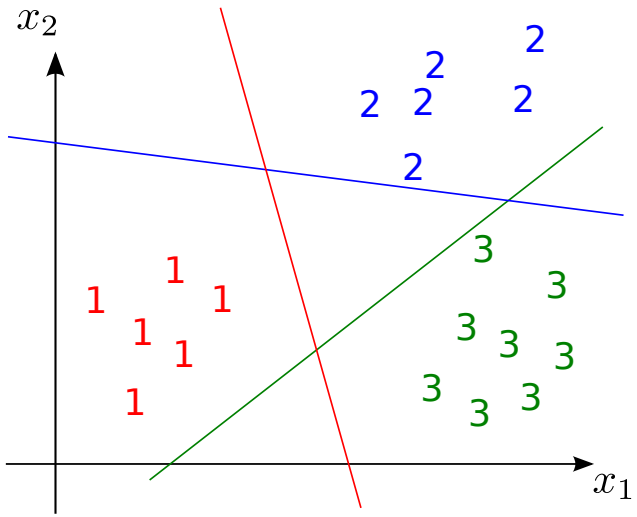


Figure: K one-vs-all classifiers

Q: *Any potential problems with training K one-vs-all classifiers?*

One-vs-rest (one-vs-all)

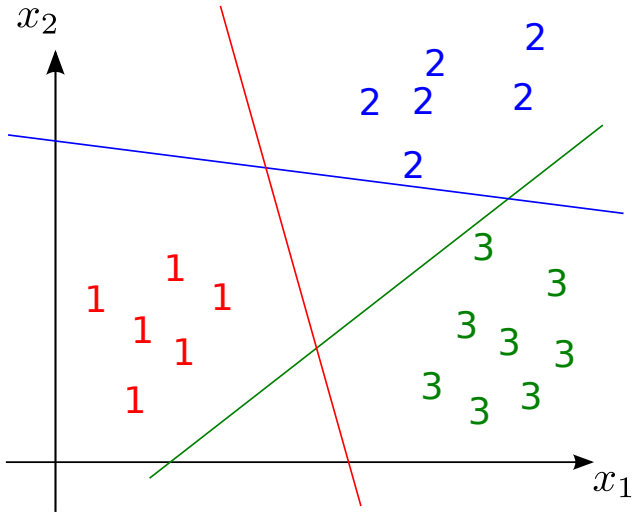


Figure: K one-vs-all classifiers

- ambiguous regions
- needs heuristic to disambiguate (e.g. $\operatorname{argmax}_k y_k(\underline{\mathbf{x}})$)
  but any heuristic needs to deal with the different scales of each $y_k(\underline{\mathbf{x}})$
- each classifier faces an imbalanced training set. But "class weighted" SVMs exist.

Actually: one-vs-rest classification for SVMs is good enough for many problems.

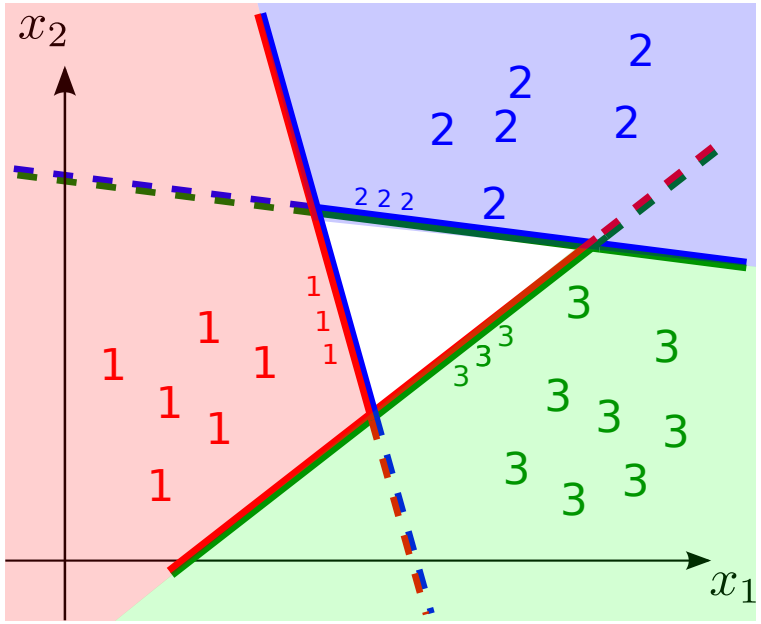
One-vs-one (all pairs)

binary classifiers $= \frac{K(K-1)}{2}$

:-(long test times.
Testing speed can be improved by organizing the pair classifiers into a DAG[1] to reduce to a subset of $K-1$ only.

:-) Training is fast as each only requires a very small subset of the data.

:-(We still have ambiguous regions:



[1] Directed Acyclic Graph, a tree would qualify.

Binary classification with error-correcting outputs

A rough outline of this ensemble method:

1. Express each class using an L-bit binary vector (e.g. $\{+1, -1\}$). Choose $L > K$.
2. Concatenate all vectors into a $K \times L$ matrix.
3. Train a binary classifier to target the value of the i-th column in this matrix.
   - If class k is associated with a "$+1$" at the i-th column. All data from this class will be used as **positive** examples when training the i-th classifier.
   - If class k is associated with a "-1" at the i-th column. All data from this class will be used as **negative** examples when training the i-th classifier.

Example for error correcting codes

Class	l_1	l_2	l_3	l_4	l_5	l_6
c_1	1	-1	1	-1	-1	1
c_2	1	1	-1	-1	1	-1
c_3	-1	1	-1	1	-1	1
c_4	-1	-1	1	-1	1	1

Table: Example for error correcting codes using 6-bits for 4 classes

Classifier $y_2(\underline{\mathbf{x}})$ will use all points from c_2, c_3 as positive examples and c_1, c_4 as negative examples.
By looking at the predictions of all classifiers. One would match the 6-bit prediction pattern with those of the 4 classes to determine which class was detected.

Support Vector Regression (SVR)

SVMs for regression problems

The regression setting

Data:

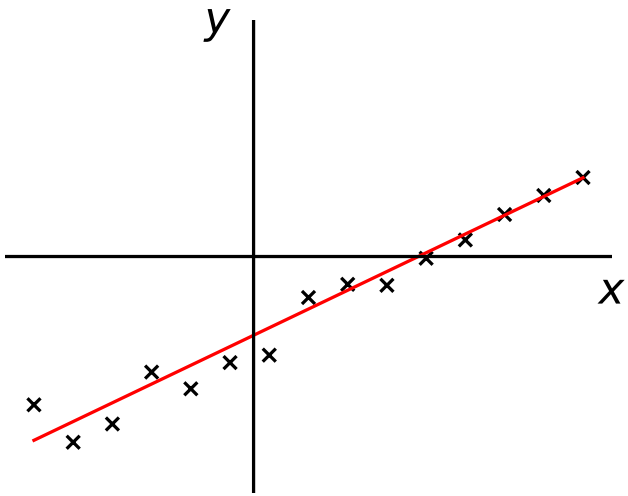
$\left\{\left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)}\right)\right\}_{\alpha=1}^{p}$ with $\underline{\mathbf{x}} \in \mathbb{R}^N$ and $y_T \in \mathbb{R}$

Model:

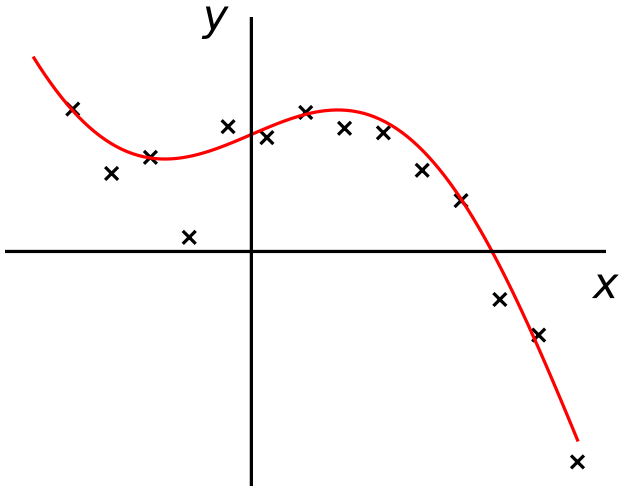
linear neuron: $y(\underline{\mathbf{x}}; \underline{\mathbf{w}}, b) = \underline{\mathbf{w}}^\top \underline{\mathbf{x}} + b$

Objective:

- ERM: $\min_{\underline{\mathbf{w}}, b} E^T$
- SRM: $\min_{\underline{\mathbf{w}}, b} \frac{1}{2}\|\underline{\mathbf{w}}\|^2 + C R_{\text{emp}}(\underline{\mathbf{w}}, b)$



(a) linear regression

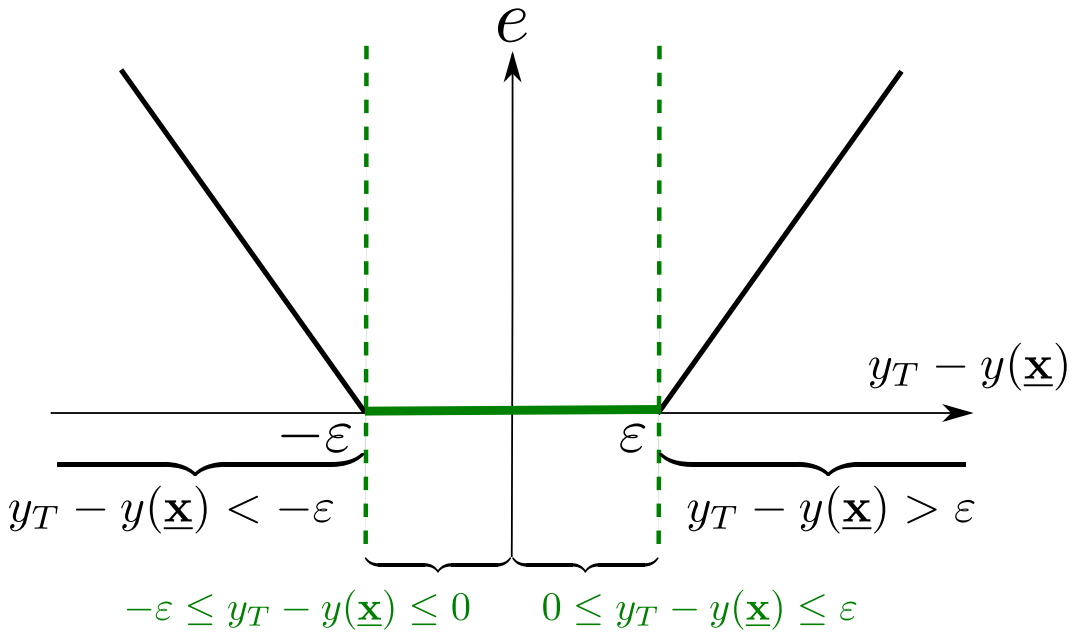


(b) non-linear regression

The ε-insensitive cost function for regression

$$e(y(\underline{\mathbf{x}};\underline{\mathbf{w}},b),y_T;\varepsilon) = e(\underbrace{y(\underline{\mathbf{x}})}_{\text{for brevity}},y_T;\varepsilon) = \max(\ 0\ ,\ \left|y(\underline{\mathbf{x}}) - y_T\right| - \varepsilon\ )$$

- The cost function is parameterized by ε that is chosen a priori.



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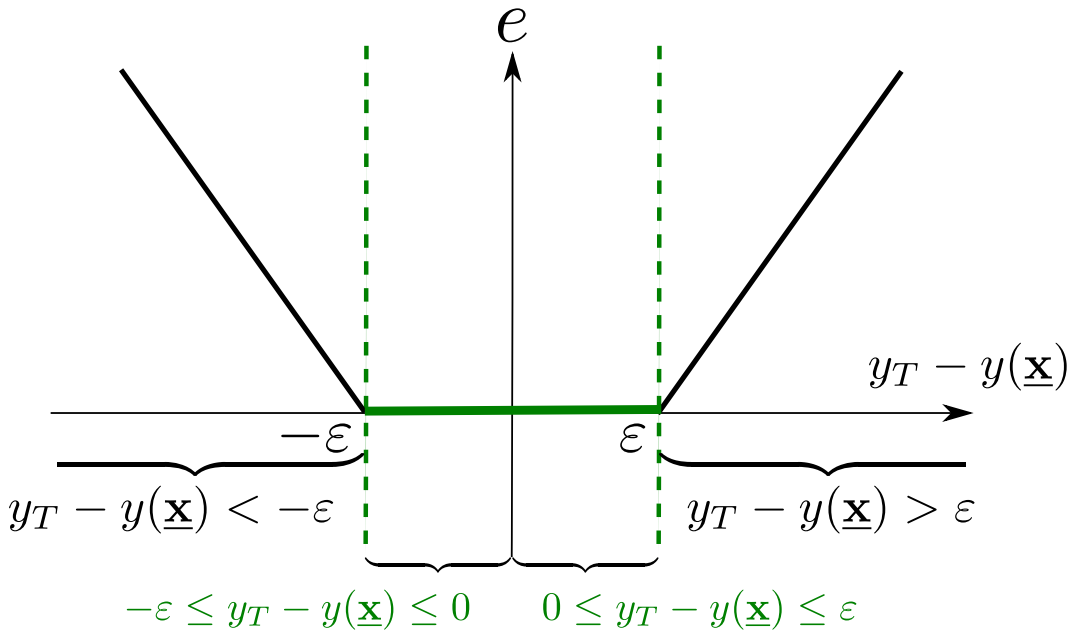


The SRM objective becomes:

$$\min_{\underline{\mathbf{w}},b} \frac{1}{2}\|\underline{\mathbf{w}}\|^2 + C \cdot \frac{1}{p}\sum_{\alpha=1}^{p} e(y(\underline{\mathbf{x}}),y_T;\varepsilon)$$

ε-SVR

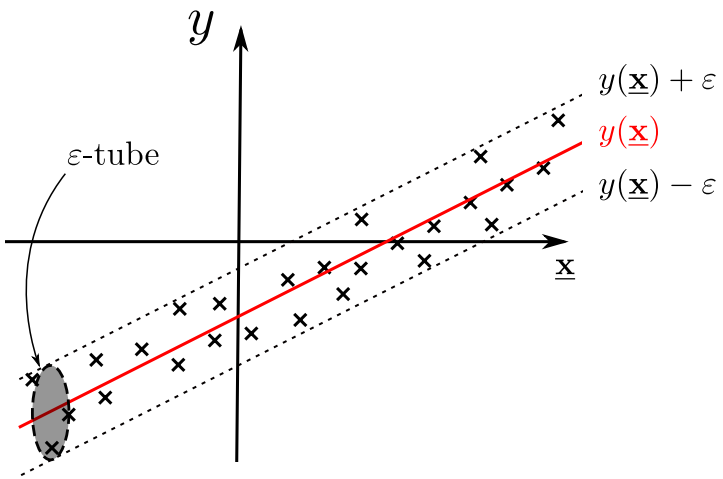


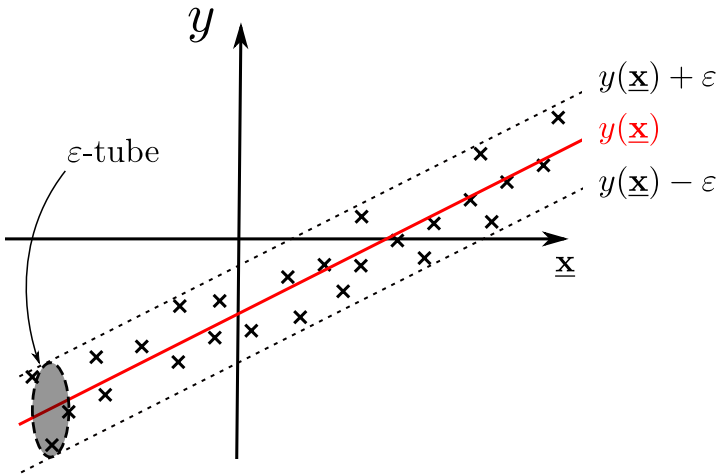
Figure: Linear model (no points outside the margin)

(i) Simplify assumptions (will be relaxed later)[2]

1. perfect regression solution, i.e. zero error ε-insensitive cost,

[2] This is based on my understanding of Ch. 9 from (Scholkopf and Smola, 2001).

ε-SVR



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1. perfect regression solution, i.e. zero error ε-insensitive cost,
2. the constrained regression problem

$$\begin{aligned} \min_{\underline{\mathbf{w}},b} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} \quad & y(\underline{\mathbf{x}}^{(\alpha)}) - y_T^{(\alpha)} \leq \varepsilon \\ & y_T^{(\alpha)} - y(\underline{\mathbf{x}}^{(\alpha)}) \leq \varepsilon \quad \text{for } \alpha = 1, \ldots, p \end{aligned}$$

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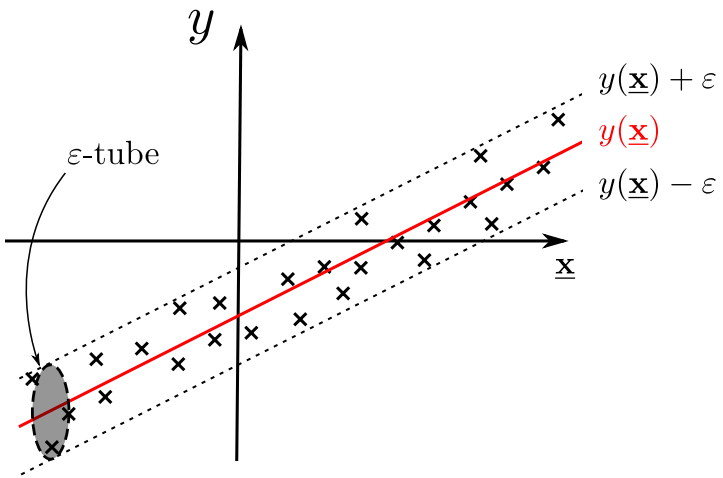
ε-SVR



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   Substituting $y(\underline{\mathbf{x}}^{(\alpha)}) = \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b$ yields:

$$\begin{aligned} \min_{\underline{\mathbf{w}},b} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\ \text{subject to} \quad & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon \\ & y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon \quad \text{for } \alpha = 1, \ldots, p \end{aligned}$$

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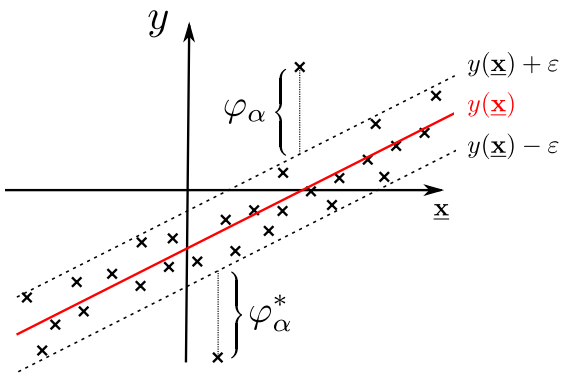
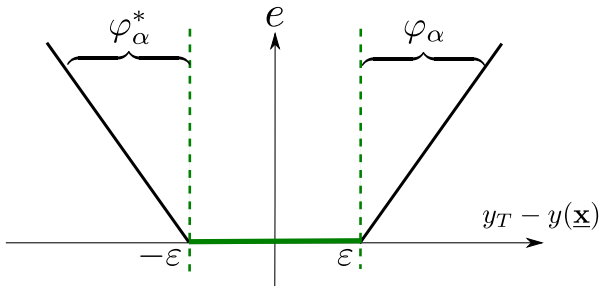
ε-SVR requires slack variables

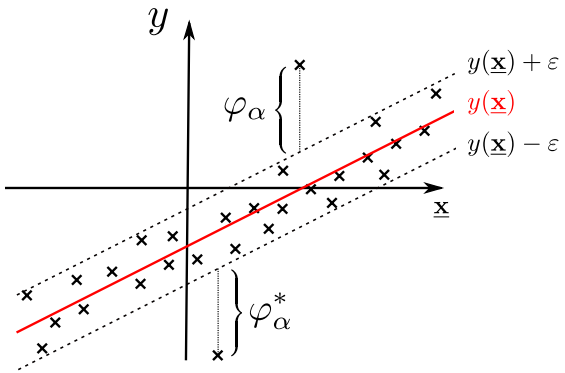


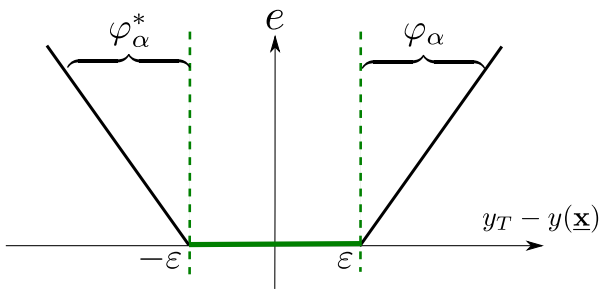


(ii) Relax constraints by allowing for errors

Q: *Why would we ever want to allow for errors?*

ε-SVR requires slack variables





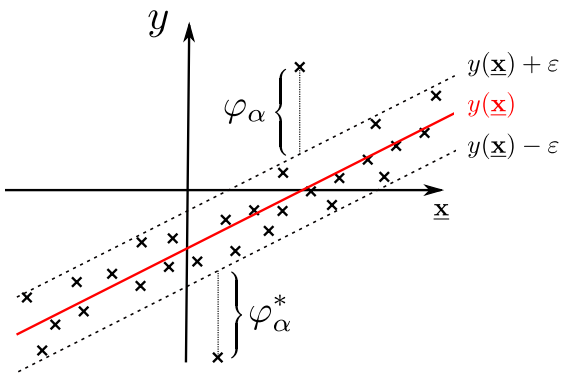
(ii) Relax constraints by allowing for errors

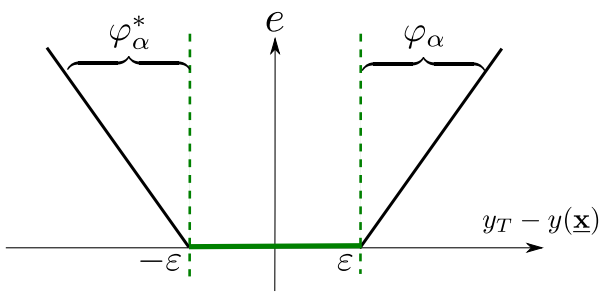
1. possibly more robust to noise and outliers,
2. constrained regression problem (with slack variables φ_α)

$$
\begin{aligned}
&\min_{\underline{\mathbf{w}},b} && \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 \\
&\text{subject to} && \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha \\
& && y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi^*_\alpha \\
& && \varphi_\alpha, \varphi^*_\alpha \geq 0 \qquad\qquad \text{for } \alpha = 1, \ldots, p
\end{aligned}
$$

Q: *How do we prevent solutions with arbitrarily large $\varphi_\alpha, \varphi^*_\alpha$ that prevent any violation of the constraints?*

ε-SVR requires slack variables





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&\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\}} && \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\tfrac{1}{p}\textstyle\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha) \\
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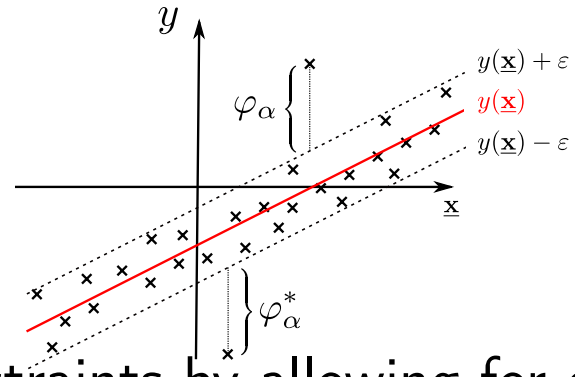
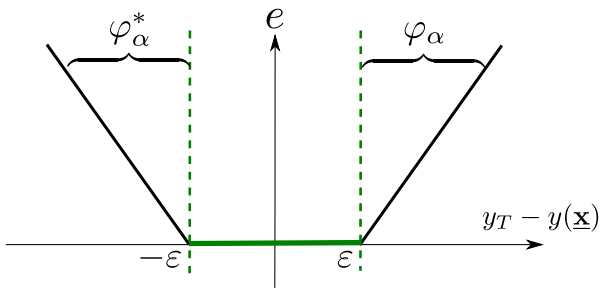
\end{aligned}
$$

with $C > 0$ (C penalizes model error)

ε-SVR requires slack variables





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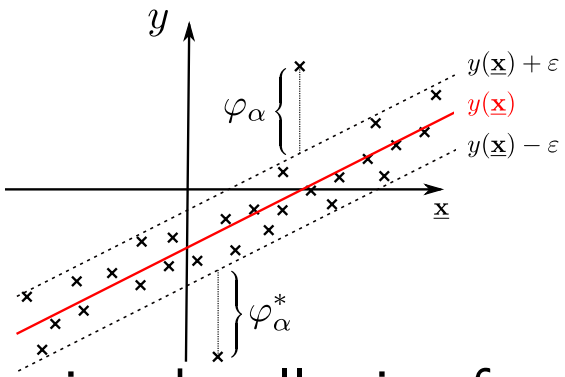
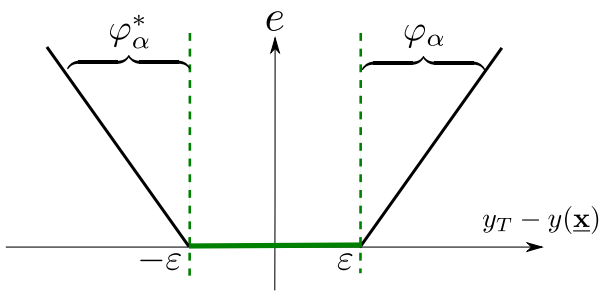
\end{aligned}
$$

with $C > 0$ (C penalizes model error)

Q: *Should we divide the sum by 2 since we're adding* $(\varphi_\alpha + \varphi^*_\alpha)$ *inside the sum?*

ε-SVR requires slack variables





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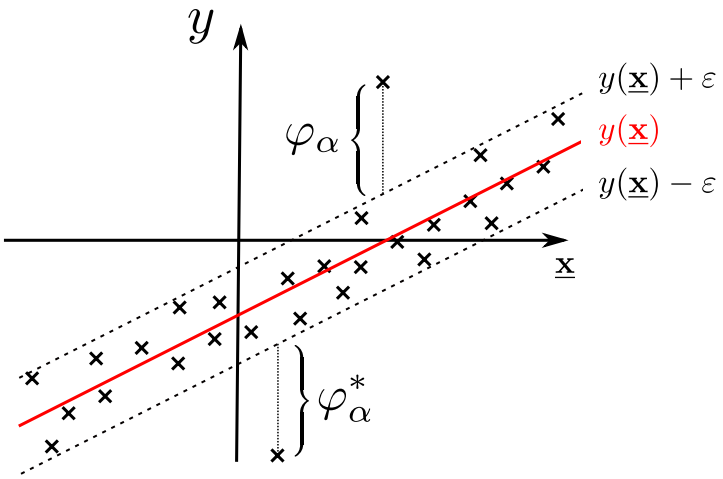
\end{aligned}$$

with $C > 0$ (C penalizes model error)

$$\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\}} \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C \overbrace{\frac{1}{p}\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha)}^{=:R^{\varepsilon}_{\mathbf{emp}}(\underline{\mathbf{w}},b)}$$

Making predictions using ε-SVR

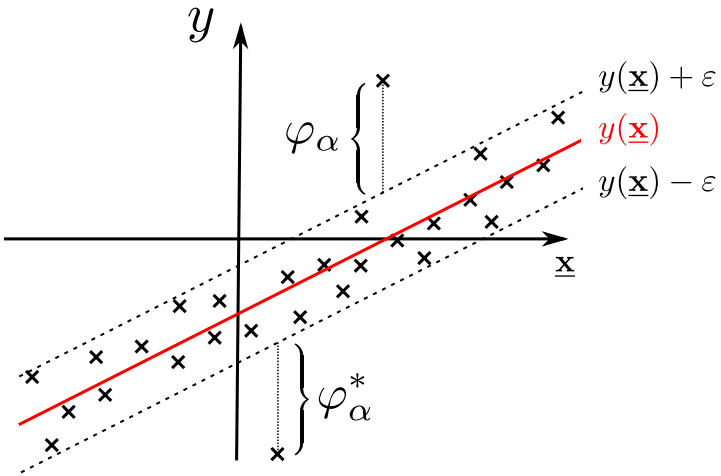


(a) linear regression

$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} (\lambda_\alpha - \lambda_\alpha^*) \underline{\mathbf{x}}^{(\alpha)} \overset{\substack{\text{either} \\ \lambda_\alpha \text{ or } \lambda_\alpha^* \neq 0 \\ \text{for SVs}}}{=} \sum_{\beta \in \mathrm{SV}} (\lambda_\beta - \lambda_\beta^*) \underline{\mathbf{x}}^{(\beta)}$$

$$y(\underline{\mathbf{x}}) = b^* + \sum_{\beta \in \mathrm{SV}} (\lambda_\beta - \lambda_\beta^*) \left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}$$

Making predictions using ε-SVR



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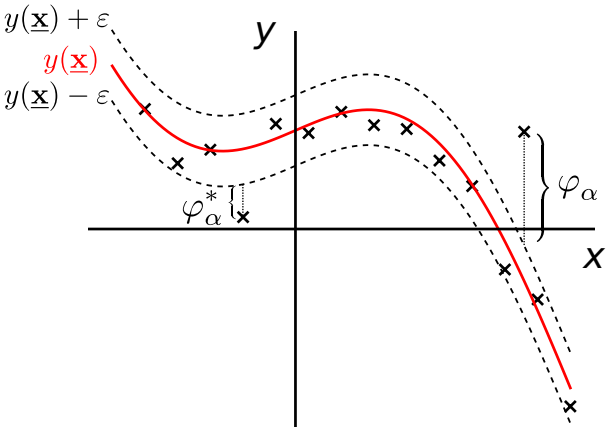


(b) non-linear regression

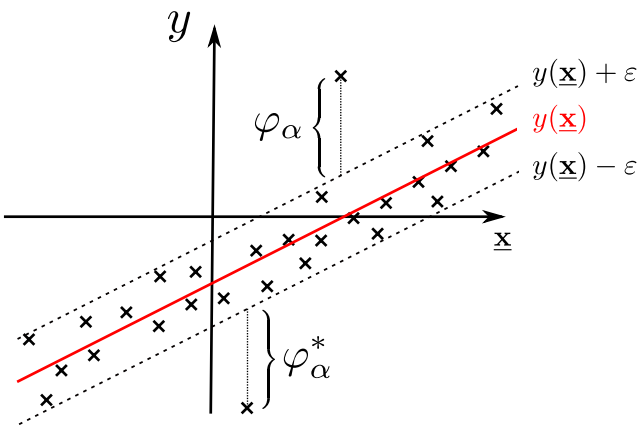
$$\underline{\mathbf{w}}^* = \sum_{\alpha=1}^{p} (\lambda_\alpha - \lambda_\alpha^*) \underline{\mathbf{x}}^{(\alpha)} \overset{\substack{\text{either} \\ \lambda_\alpha \text{ or } \lambda_\alpha^* \neq 0 \\ \text{for SVs}}}{=} \sum_{\beta \in \mathrm{SV}} (\lambda_\beta - \lambda_\beta^*) \underline{\mathbf{x}}^{(\beta)}$$

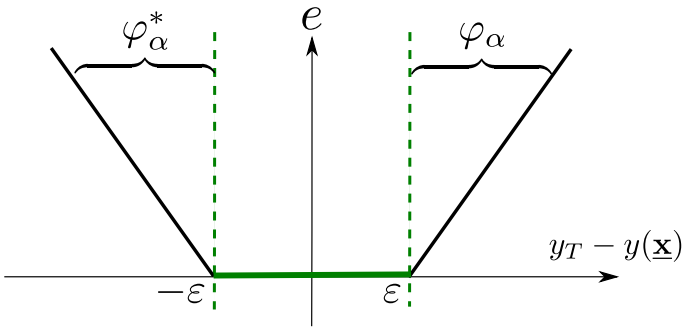
$$y(\underline{\mathbf{x}}) = b^* + \sum_{\beta \in \mathrm{SV}} (\lambda_\beta - \lambda_\beta^*) \underbrace{\left(\underline{\mathbf{x}}^{(\beta)}\right)^\top \underline{\mathbf{x}}}_{\text{kernels!}}$$

ν-SVR

Essentially ε-SVR but treat ε as a primal variable instead of a hyperparameter.

Motivation for ν-SVR





ε-SVR requires treating ε as a hyperparameter.
ν-SVR extends ε-SVR, by allowing the width of the "ε-tube" to adapt to the data. This is done by adding ε to our set of *primal variables*.
The SRM objective for ν-SVR becomes:

$$\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \frac{1}{p}\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha)\right]$$

Deriving the primal problem for ν-SVR

$$
\begin{aligned}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\}} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\tfrac{1}{p}\textstyle\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha) \\
\text{subject to} \quad & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi^*_\alpha \\
& \varphi_\alpha, \varphi^*_\alpha \geq 0 \qquad\qquad \text{for } \alpha = 1, \ldots, p
\end{aligned}
$$

with $C > 0$ (C penalizes model error)

[3] If interested, see details in Ch. 9.3 from (Scholkopf and Smola, 2001) and the paper (Schölkopf et al., 2000)

Deriving the primal problem for ν-SVR

$$
\begin{aligned}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \tfrac{1}{p}\textstyle\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha)\right] \\
\text{subject to} \quad & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi^*_\alpha \\
& \varphi_\alpha, \varphi^*_\alpha \geq 0 \qquad\qquad \text{for } \alpha = 1, \ldots, p \\
& \varepsilon \geq 0
\end{aligned}
$$

with $C > 0$ (C penalizes model error), $\nu \geq 0$

Q: *Aren't we just exchanging one hyperparameter for another? How is this better?*

[3] If interested, see details in Ch. 9.3 from (Scholkopf and Smola, 2001) and the paper (Schölkopf et al., 2000)

Deriving the primal problem for ν-SVR

$$
\begin{aligned}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \tfrac{1}{p}\textstyle\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha)\right] \\
\text{subject to} \quad & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi^*_\alpha \\
& \varphi_\alpha, \varphi^*_\alpha \geq 0 \qquad\qquad \text{for } \alpha = 1, \ldots, p \\
& \varepsilon \geq 0
\end{aligned}
$$

with $C > 0$ (C penalizes model error), $\nu \geq 0$

Q: *Aren't we just exchanging one hyperparameter for another? How is this better?*

- Yes, but with increased resolution of how we balance model error and model complexity. ν acts as a

- upper bound on the fraction of errors: $\nu \geq \frac{1}{p}\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi^*_\alpha)$
- lower bound on the fraction of SVs: $\nu \leq \frac{\#SV}{p}$[3].

[3] If interested, see details in Ch. 9.3 from (Scholkopf and Smola, 2001) and the paper (Schölkopf et al., 2000)

Deriving the Lagrangian for the primal problem of ν-SVR
Identifying the Lagrange multipliers

Q: *Where are multipliers needed?*

The primal problem

$$
\begin{aligned}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi_\alpha^*\},\varepsilon} \quad & \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \tfrac{1}{p}\textstyle\sum_\alpha^p(\varphi_\alpha + \varphi_\alpha^*)\right] \\
\text{subject to} \quad & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi_\alpha^* \\
& \varphi_\alpha \geq 0 \\
& \varphi_\alpha^* \geq 0 \qquad\qquad \text{for } \alpha = 1, \ldots, p \\
& \varepsilon \geq 0
\end{aligned}
$$

with $C > 0$ and $\nu \geq 0$.

Deriving the Lagrangian for the primal problem of ν-SVR
Identifying the Lagrange multipliers

Q: *Where are multipliers needed?*

The primal problem

$$
\begin{array}{lll}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi_\alpha^*\},\varepsilon} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \frac{1}{p}\sum_\alpha^p(\varphi_\alpha + \varphi_\alpha^*)\right] & \\
\text{subject to} & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha & \rightsquigarrow \{\lambda_\alpha\} \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi_\alpha^* & \rightsquigarrow \{\lambda_\alpha^*\} \\
& \varphi_\alpha \geq 0 & \rightsquigarrow \{\eta_\alpha\} \\
& \varphi_\alpha^* \geq 0 \qquad \text{for } \alpha = 1,\ldots,p & \rightsquigarrow \{\eta_\alpha^*\} \\
& \varepsilon \geq 0 & \rightsquigarrow \delta
\end{array}
$$

with $C > 0$ and $\nu \geq 0$.

Next: re-arrange the constraints to be in the form $f_\alpha \leq 0$ to form the Lagrangian of the primal problem.

Deriving the Lagrangian for the primal problem of ν-SVR
Re-arranging the constraints

Next: re-arrange the constraints to be in the form $f_\alpha \leq 0$:

The primal problem

$$
\begin{array}{lll}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \frac{1}{p}\sum_\alpha^p(\varphi_\alpha + \varphi^*_\alpha)\right] & \\
\text{subject to} & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} \leq \varepsilon + \varphi_\alpha & \rightsquigarrow \{\lambda_\alpha\} \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b \leq \varepsilon + \varphi^*_\alpha & \rightsquigarrow \{\lambda^*_\alpha\} \\
& \varphi_\alpha \geq 0 & \rightsquigarrow \{\eta_\alpha\} \\
& \varphi^*_\alpha \geq 0 \qquad \text{for } \alpha = 1,\ldots,p & \rightsquigarrow \{\eta^*_\alpha\} \\
& \varepsilon \geq 0 & \rightsquigarrow \delta
\end{array}
$$

with $C > 0$ and $\nu \geq 0$.

Deriving the Lagrangian for the primal problem of ν-SVR
Re-arranging the constraints

The primal problem with constraints in the form of $f_\alpha \leq 0$

$$
\begin{array}{lll}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \frac{1}{p}\sum_\alpha^p(\varphi_\alpha + \varphi^*_\alpha)\right] & \\
\text{subject to} & \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b - y_T^{(\alpha)} - \varepsilon - \varphi_\alpha \leq 0 & \rightsquigarrow \{\lambda_\alpha\} \\
& y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b - \varepsilon - \varphi^*_\alpha \leq 0 & \rightsquigarrow \{\lambda^*_\alpha\} \\
& -\varphi_\alpha \leq 0 & \rightsquigarrow \{\eta_\alpha\} \\
& -\varphi^*_\alpha \leq 0 \qquad \text{for } \alpha = 1,\ldots,p & \rightsquigarrow \{\eta^*_\alpha\} \\
& -\varepsilon \leq 0 & \rightsquigarrow \delta
\end{array}
$$

with $C > 0$ and $\nu \geq 0$.

Deriving the Lagrangian for the primal problem of ν-SVR
Re-arranging the constraints

The primal problem with constraints in the form of $f_\alpha \leq 0$

$$
\begin{array}{llll}
\min_{\underline{\mathbf{w}},b,\{\varphi_\alpha\},\{\varphi^*_\alpha\},\varepsilon} & & \frac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\left[\nu\varepsilon + \frac{1}{p}\sum_\alpha^p(\varphi_\alpha + \varphi^*_\alpha)\right] & \\
\text{subject to} & - & \{\varphi_\alpha + \varepsilon + y_T^{(\alpha)} - \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} - b\} \leq 0 & \rightsquigarrow \{\lambda_\alpha\} \\
& - & \{\varphi^*_\alpha + \varepsilon - y_T^{(\alpha)} + \underline{\mathbf{w}}^\top \underline{\mathbf{x}}^{(\alpha)} + b\} \leq 0 & \rightsquigarrow \{\lambda^*_\alpha\} \\
& - & \varphi_\alpha \leq 0 & \rightsquigarrow \{\eta_\alpha\} \\
& - & \varphi^*_\alpha \leq 0 \qquad \text{for } \alpha = 1,\ldots,p & \rightsquigarrow \{\eta^*_\alpha\} \\
& - & \varepsilon \leq 0 & \rightsquigarrow \delta
\end{array}
$$

with $C > 0$ and $\nu \geq 0$.

Deriving the Lagrangian for the primal problem of ν-SVR

$$L(\underbrace{\underline{\mathbf{w}}, b, \{\varphi_\alpha\}, \{\varphi_\alpha^*\}, \varepsilon}_{\text{primal variables}}, \underbrace{\{\lambda_\alpha\}, \{\lambda_\alpha^*\}, \{\eta_\alpha\}, \{\eta_\alpha^*\}, \delta}_{\text{dual variables (Lagrange multipliers)}})$$

$$= \tfrac{1}{2}\|\underline{\mathbf{w}}\|_2^2 + C\Big[\nu\varepsilon + \tfrac{1}{p}\sum_{\alpha=1}^{p}(\varphi_\alpha + \varphi_\alpha^*)\Big]$$

$$-\sum_{\alpha=1}^{p}\lambda_\alpha\{\varphi_\alpha + \varepsilon + y_T^{(\alpha)} - \underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)} - b\}$$

$$-\sum_{\alpha=1}^{p}\lambda_\alpha^*\{\varphi_\alpha^* + \varepsilon - y_T^{(\alpha)} + \underline{\mathbf{w}}^\top\underline{\mathbf{x}}^{(\alpha)} + b\}$$

$$-\sum_{\alpha=1}^{p}\eta_\alpha\varphi_\alpha - \sum_{\alpha=1}^{p}\eta_\alpha^*\varphi_\alpha^* - \delta\varepsilon$$

with $\{\lambda_\alpha\}, \{\lambda_\alpha^*\}, \{\eta_\alpha\}, \{\eta_\alpha^*\}, \delta \geq 0$

Optimizing ν-SVR

1. We start with optimizing the Lagrangian of the *primal problem*. As we did with SVMs and C-SVMs, this is achieved by
   a. calculating the derivatives of L w.r.t. *primal variables* $\underline{\mathbf{w}}, b, \{\varphi_\alpha\}, \{\varphi_\alpha^*\}, \varepsilon$
   b. setting the partial derivatives to zero. (**see homework**)
2. Substituting the solutions to the *primal problem* back into the Lagrangian yields the Lagrangian of the *dual problem*:

$$\max_{\{\lambda_\alpha\},\{\lambda_\alpha^*\}} -\frac{1}{2}\sum_{\alpha,\beta=1}^{p}(\lambda_\alpha^*-\lambda_\alpha)(\lambda_\beta^*-\lambda_\beta)(\underline{\mathbf{x}}^{(\alpha)})^\top\underline{\mathbf{x}}^{(\beta)}+\sum_{\alpha=1}^{p}(\lambda_\alpha^*-\lambda_\alpha)y_T^{(\alpha)}$$

s.t. $\forall\alpha\in\{1,\ldots,p\}$:

$$0\le\lambda_\alpha\le\frac{C}{p}\,,\quad 0\le\lambda_\alpha^*\le\frac{C}{p}\,,\quad \sum_{\alpha=1}^{p}(\lambda_\alpha-\lambda_\alpha^*)=0\,,\quad \sum_{\alpha=1}^{p}(\lambda_\alpha+\lambda_\alpha^*)\le\nu C\,.$$

References I

Bernhard Scholkopf and Alexander J Smola. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. MIT press, 2001.

Bernhard Schölkopf, Alex J Smola, Robert C Williamson, and Peter L Bartlett. New support vector algorithms. *Neural computation*, 12(5):1207–1245, 2000. URL https://alex.smola.org/papers/2000/SchSmoWilBar00.pdf.